Measures of spread

1 For each of the following sets of scores:

- **i** Sort the scores in ascending order.
- **ii** Find the number of scores.
- **iii** Find the median.
- **iv** Find the lower quartile of the set of scores.
- **v** Find the upper quartile of the set of scores.
- **vi** Find the interquartile range.

a $40, 39, 15, 17, 10, 6, 24$

b $-4, -6, -1, 7, 9, 7, 9$

c $8, 20, 19, 4, 15, 14, 10$

d $42, 28, 22, 40, 20, 54, 32, 43$

e $84, 85, 79, 71, 69, 88, 82, 78$

f $102, 115, 110, 113, 100$

g $228, 205, 198, 202, 207, 197$

h $19, 29, 55, 22, 15, 46, 35, 9, 27, 40$

2 Consider the following data set containing 30 scores:

$$10, 13, 15, 24, 32, 42, 46, 53, 58, 64$$

$$11, 14, 16, 27, 33, 42, 49, 53, 60, 67$$

$$11, 15, 18, 28, 37, 44, 51, 55, 61, 67$$

a Find the median.

b Find the interquartile range.

3 Find the interquartile range for the following data set containing 20 scores:

$$5, 8, 11, 12, 12, 14, 15, 15, 18, 18, 19, 21, 21, 25, 29, 32, 33, 37, 38, 38$$

4 For each of the following data sets:

i Find the total number of scores.

ii Find the median.

iii Find the lower quartile of the set of scores.

iv Find the upper quartile of the set of scores.

v Find the interquartile range.

a $33, 38, 50, 12, 33, 48, 41$

b $-3, -3, 1, 9, 9, 6, -9$

c

	Leaf
2	2 5 6 7 9
3	0 0 5 6 8
4	0 0 1 8 9

Key: $1|2 = 12$

d

	Leaf
2	1 3 4 6 9
3	1 2 2 2 6
4	2 3 5 6 7

Key: $1|2 = 12$

e

Score	Frequency
5	3
13	3
16	2
28	2
31	3
38	4
48	2

f

Score	Frequency
5	1
14	1
18	3
24	2
32	1
38	2
50	5

5 If there are 78 scores in a set of data, in which position will the lower quartile lie?

6 The following list shows the number of points scored by a basketball team in each game of their previous season:

$$59,\ 67,\ 73,\ 82,\ 91,\ 58,\ 79,\ 88,\ 69,\ 84,\ 55,\ 80,\ 98,\ 64,\ 82$$

a State the maximum value.

b State the minimum value.

c Find the median value.

d Find the lower quartile score.

e Find the upper quartile score.

7 The following data set shows Ray's scores [redacted] her last 13 rounds of golf played:

$$66, 66, 68, 68, 70, 78, 80, 84, 106, 116, 126, 130, 132$$

a Find her median score.

b Find the lower quartile score.

c Find the upper quartile score.

d Find the interquartile range.

8 The following data set shows Luke's scores from his last 17 exams:

$$42, 46, 48, 51, 52, 54, 56, 68, 72, 76, 78, 82, 85, 86, 88, 92, 96$$

a Find his median score.

b Find the lower quartile score.

c Find the upper quartile score.

d Find the interquartile range.

9 There is a test to measure the Emotional Quotient (EQ) of an individual. Here are the EQ results for 21 people listed in ascending order:

$$90,\ 90,\ 91,\ 92,\ 93,\ 94,\ 95,\ 95,\ 95,\ 97,\ 99,\ 100,\ 108,\ 114,\ 116,\ 116,\ 117,\ 118,\ 118,\ 122,\ 129$$

a Find the median EQ score.

b Find the upper quartile score.

c Find the lower quartile score.

d Find the interquartile range.

10 In a competition, a contestant must complete 12 challenges earning as many points as possible. Her scores for the first 11 challenges are:

$$38, 43, 45, 66, 67, 82, 92, 102, 105, 108, 119$$

Determine her score in the 12th round if the lower quartile of all of her 12 scores is 55.

11 The stem-and-leaf plot shows the number of hours students spent studying during an entire semester:

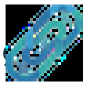
a Find the lower quartile of the set of scores.

b Find the upper quartile of the set of scores.

c Find the interquartile range.

	Leaf
6	2 7
7	1 2 2 4 7 9
8	0 1 2 5 7
9	0 1

Key: $5 \mid 2 = 52$

12 Below is the luggage weight of 30 passen

a Find the mean luggage weight to two decimal places.

b Determine the following in kilograms:

i Median

ii Lower Quartile

iii Upper Quartile

c In which quartile does the mean lie?

Weight	Frequency
16	4
17	5
18	5
19	3
20	4
21	6
22	3

13 A group of students were asked how many phone calls they had made the previous day. The information was collected in the following frequency table:

a How many students were surveyed?

b Find the range of the data.

c Find the interquartile range.

Phone calls	Frequency
0	8
1	5
2	10
3	6
4	6
5	7
6	3
7	2

14 A sample of boxes of matches were selected for quality control and the number of matches in each box recorded in the given frequency table:

a How many matchboxes were sampled?

b Find the range of the data.

c Find the interquartile range.

Matches	Frequency
45	1
46	2
47	5
48	3
49	7
50	25
51	6
52	2

15 A group of students were asked how many phone calls they had made the previous day. The information was collected in the given frequency table:

Phone calls	Frequency
0	8
1	5
2	10
3	6
4	6
5	7
6	3
7	2

a How many students were surveyed?

b Find the range of the data.

c Find the interquartile range.

16 Consider the following bar chart:

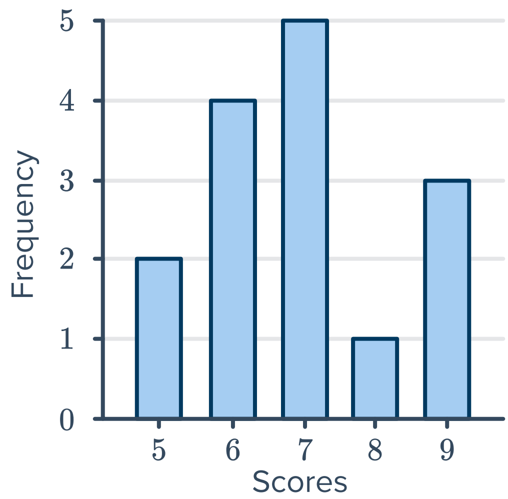
a Organise the data into a frequency table.

b Find the median score using the distribution table.

c Find the lower quartile score.

d Find the upper quartile score.

e Find the interquartile range.

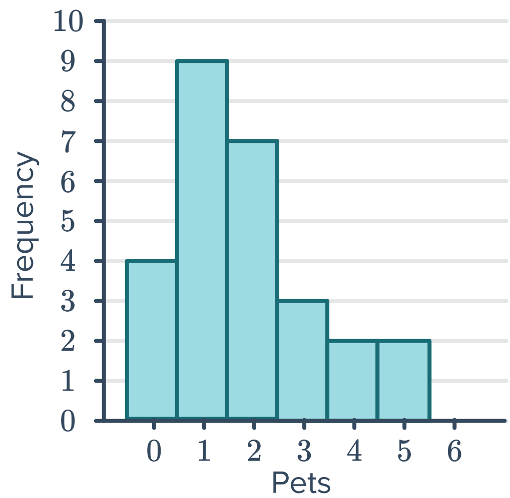


17 The column graph shows the number of pets that each student in a class owns:

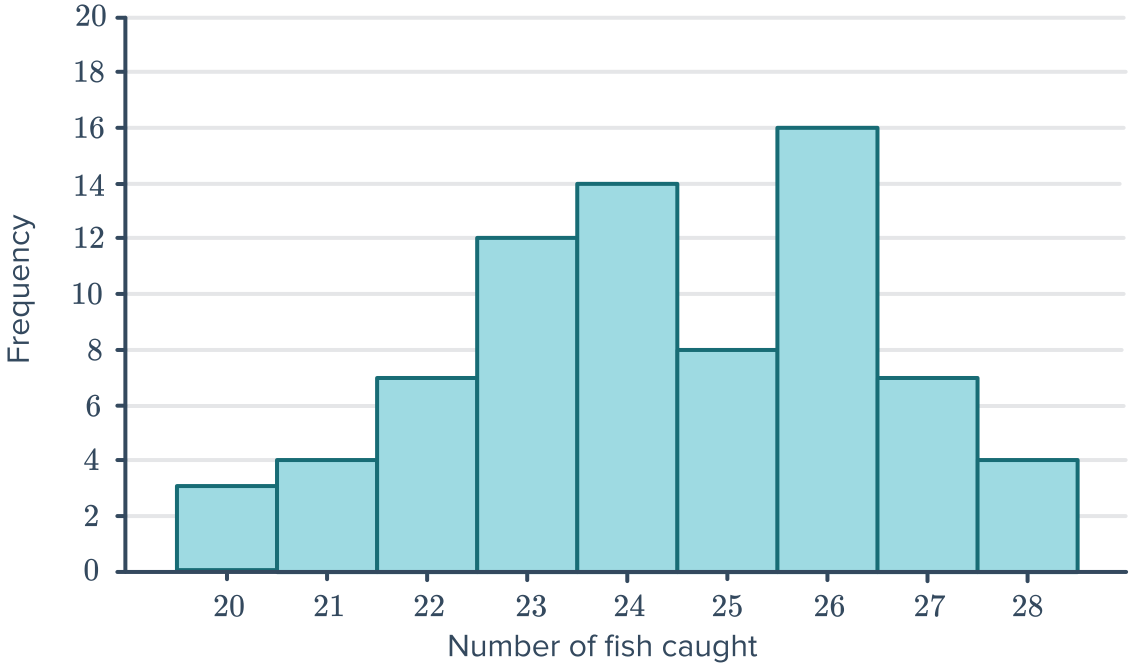
a Find the lower quartile of the set of scores.

b Find the upper quartile of the set of scores.

c Find the interquartile range.



18 Dylan records the number of fish he catch[redacted] each fishing trip over a period of time:



a How many fishing trips did he go on?

b Find the range in the number of fish caught.

c Find the interquartile range.

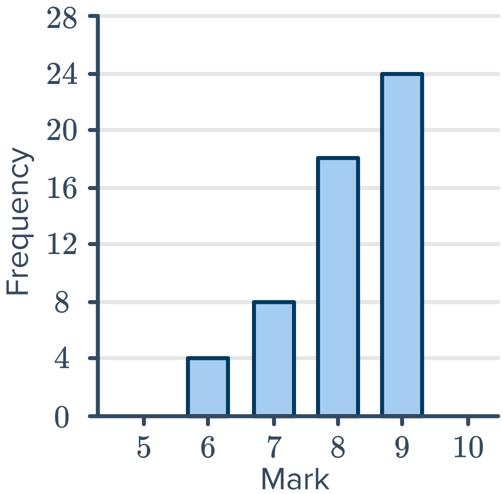
19 The bar graph shows the marks (out of 10) that students received on a spelling test:

a Find the lower quartile of the set of scores.

b Find the upper quartile of the set of scores.

c Find the interquartile range.

d Explain the effect the scores less than 7 have on the summary statistics.



20 The stem-and-leaf plot shows the batting [redacted]s of two cricket teams, A and B:

a What is the median score of Team A?

b What is the median score of Team B?

c What is the range of Team A's scores?

d What is the range of Team B's scores?

e What is the interquartile range of Team A's scores?

f What is the interquartile range of Team B's scores?

g Explain the differences between Team A and Team B's distribution of scores.

Plot title

Team A		Team B
6 5 4	6	0 5 8
8 5 4 0	7	1 5 7
8 6	8	2 3 7 9
	9	2 5

Key: $6 \mid 1 \mid 2 =$ 12 and 16

21 To gain a place in the main race of a car rally, teams must compete in a qualifying round. The median time in the qualifying round determines the cut off time to make it through to the main race. Below are some results from the qualifying round:

- 75% of teams finished in 159 minutes or less.
- 25% of teams finished in 132 minutes or less.
- 25% of teams finished between with a time between 132 and 142 minutes.

a Find the cut off time required in the qualifying round to make it through to the main race.

b Find the interquartile range in the qualifying round.

c In the qualifying round, the ground was wet, while in the main race, the ground was dry. To make the times more comparable, the finishing time of each team from the qualifying round is reduced by 5 minutes.

Find the new median time from the qualifying round.